Reinforcement Learning: Mathematical Foundations

Lecture Notes

2021

1 Introduction

Definition 1.1 (Intelligence)**.** *Intelligence is the ability to learn to make decisions to achieve goals.*

Definition 1.2 (Reinforcement Learning)**.** *Reinforcement learning is the science and framework of learning to make decisions from interaction with an environment.*

1.1 The Reward Hypothesis

Theorem 1.1 (Reward Hypothesis)**.** *Any goal can be formalized as the outcome of maximizing a cumulative reward.*

2 Formalization of the RL Problem

2.1 Agent-Environment Interaction

Definition 2.1 (History)**.** *The history is the sequence of observations, actions, and rewards up to time t:*

$$\mathcal{H}_t = O_0, A_0, R_1, O_1, \ldots, O_{t-1}, A_{t-1}, R_t, O_t$$

Definition 2.2 (Agent State)**.** *The agent state S_t is a function of the history:*

$$S_t = u(S_{t-1}, A_{t-1}, R_t, O_t)$$

where u is the state update function.

2.2 Rewards and Returns

Definition 2.3 (Reward)**.** *A reward $R_t \in \mathbb{R}$ is a scalar feedback signal at time t indicating how well the agent is performing.*

Definition 2.4 (Return). *The return G_t is the cumulative discounted reward from time t:*

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \cdots = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$$

where $\gamma \in [0, 1]$ is the discount factor.

Proposition 2.1 (Recursive Form of Return). *The return satisfies the recursive relation:*

$$G_t = R_{t+1} + \gamma G_{t+1}$$

2.3 Value Functions

Definition 2.5 (State-Value Function). *For a policy π, the state-value function $v_\pi : \mathcal{S} \to \mathbb{R}$ is defined as:*

$$v_\pi(s) = \mathbb{E}[G_t \mid S_t = s, \pi] = \mathbb{E}\left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \mid S_t = s, \pi\right]$$

Definition 2.6 (Action-Value Function). *For a policy π, the action-value function $q_\pi : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ is defined as:*

$$q_\pi(s, a) = \mathbb{E}[G_t \mid S_t = s, A_t = a, \pi] = \mathbb{E}\left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \mid S_t = s, A_t = a, \pi\right]$$

Theorem 2.1 (Bellman Equation for v_π). *The state-value function satisfies the Bellman equation:*

$$v_\pi(s) = \mathbb{E}[R_{t+1} + \gamma v_\pi(S_{t+1}) \mid S_t = s, A_t \sim \pi(s)]$$

Definition 2.7 (Optimal Value Function). *The optimal state-value function is:*

$$v_*(s) = \max_\pi v_\pi(s)$$

Theorem 2.2 (Bellman Optimality Equation). *The optimal value function satisfies:*

$$v_*(s) = \max_a \mathbb{E}[R_{t+1} + \gamma v_*(S_{t+1}) \mid S_t = s, A_t = a]$$

2.4 Markov Decision Processes

Definition 2.8 (Markov Property). *A decision process is Markov if:*

$$p(r, s' \mid S_t, A_t) = p(r, s' \mid \mathcal{H}_t, A_t)$$

That is, the state contains all relevant information from the history.

Definition 2.9 (Markov Decision Process (MDP))**.** *A Markov Decision Process is a tuple* $(\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma)$ *where:*

- $\mathcal{S}$ *is the state space*
- $\mathcal{A}$ *is the action space*
- $\mathcal{P} : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \to [0, 1]$ *is the transition probability function*
- $\mathcal{R} : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ *is the reward function*
- $\gamma \in [0, 1]$ *is the discount factor*

Definition 2.10 (Fully Observable Environment)**.** *An environment is fully observable if* $O_t = S_t^{env}$ *where* S_t^{env} *is the environment state.*

Definition 2.11 (Partially Observable MDP (POMDP))**.** *A partially observable environment where observations are not Markovian, but the underlying environment state may be.*

2.5 Policies

Definition 2.12 (Policy)**.** *A policy is a mapping from states to actions. It can be:*

- *Deterministic:* $\pi : \mathcal{S} \to \mathcal{A}$ *where* $A = \pi(S)$
- *Stochastic:* $\pi : \mathcal{S} \times \mathcal{A} \to [0, 1]$ *where* $\pi(a|s) = p(A = a \mid S = s)$

Definition 2.13 (Optimal Policy)**.** *An optimal policy is:*

$$\pi_*(s) = \arg\max_{\pi} v_\pi(s)$$

2.6 Models

Definition 2.14 (Transition Model)**.** *A transition model predicts the next state:*

$$\mathcal{P}(s, a, s') \approx p(S_{t+1} = s' \mid S_t = s, A_t = a)$$

Definition 2.15 (Reward Model)**.** *A reward model predicts the immediate reward:*

$$\mathcal{R}(s, a) \approx \mathbb{E}[R_{t+1} \mid S_t = s, A_t = a]$$

3 Agent Components and Categories

Definition 3.1 (Agent Components)**.** *An RL agent may contain:*

- *Agent state* S_t
- *Policy* π

- *Value function v or q*
- *Model $(\mathcal{P}, \mathcal{R})$*

Definition 3.2 (Agent Categories by Components)**.** *Agents can be categorized as:*

- ***Value-based****: No explicit policy (implicit), has value function*
- ***Policy-based****: Has policy, no value function*
- ***Actor-Critic****: Has both policy and value function*

Definition 3.3 (Agent Categories by Model)**.** *Agents can be categorized as:*

- ***Model-free****: Policy and/or value function, no model*
- ***Model-based****: Has model (optionally policy and/or value function)*

4 Subproblems of Reinforcement Learning

Definition 4.1 (Prediction Problem)**.** *Given a policy π, evaluate the future by computing $v_\pi(s)$ for all $s \in \mathcal{S}$.*

Definition 4.2 (Control Problem)**.** *Find the optimal policy π_* that maximizes $v_\pi(s)$ for all $s \in \mathcal{S}$.*

Definition 4.3 (Learning)**.** *The environment is initially unknown. The agent learns by interacting with the environment.*

Definition 4.4 (Planning)**.** *A model of the environment is given (or learned). The agent computes a policy by reasoning in the model without external interaction.*

5 Examples

Example 5.1 (Maze Navigation)**.** *Consider a grid maze with:*

- *States: Agent's location (x, y)*
- *Actions: $\mathcal{A} = \{North, East, South, West\}$*
- *Reward: $R = -1$ per time-step*
- *Goal: Reach target location*

The value function $v_\pi(s)$ represents the expected number of steps to reach the goal from state s under policy π (negated).

Example 5.2 (Gridworld)**.** *A 5×5 gridworld with special states A and B that teleport to A' and B' with rewards $+10$ and $+5$ respectively.*

- *Actions:* $\{$*up, down, left, right*$\}$
- *Reward:* -1 *when bumping into walls, otherwise* 0
- *Discount:* $\gamma = 0.9$